

BROOKE KOTTEN

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EDUCATION

University of Michigan

PhD, Astronomy, expected 2029

August 2024-Present

Ann Arbor, MI

University of Wisconsin-Madison

B.S., Astronomy-Physics (Honors) and Physics

September 2021-May 2024

Madison, WI

Saint Cloud State University

A.A., Liberal Arts, *summa cum laude*

Concurrent with high school diploma

August 2019-May 2021

Saint Cloud, MN

RESEARCH EXPERIENCE

Graduate Student at the University of Michigan

August 2024-Present

with Dr. Mary Anne Limbach (University of Michigan)

Searching for transiting exosatellites in JWST time series observations of substellar worlds and investigating the detectability of exo-Ios.

Lamat REU & UW-Madison Senior Honors Thesis

June 2023-August 2024

with Professor Melinda Soares-Furtado (UW-Madison) and Professor Enrico Ramirez-Ruiz (UCSC)

Investigated the lithium enrichment in a planetary engulfment candidate.

Bolder Solar Alliance REU & Student Researcher at CIRES

May 2022-June 2023

with Dr. Courtney Peck and Dr. Janet Machol (CIRES/NOAA/NCEI)

Improved GOES-R's X-Ray Sensor (XRS) location algorithm with new calculated parameters and a correction for near-simultaneous solar flares. My location correction method is used operationally to improve the real-time flare locations from XRS.

Undergraduate Research Scholars Program

October 2021-May 2022

with Dr. Ralf Kotulla (UW-Madison)

Investigated star formation on the outskirts of spiral galaxies.

PUBLICATIONS

FIRST AUTHOR PUBLICATIONS

2. **Kotten, B.**, Limbach, M. A., Vos, J.; et al. "On the Detectability of Volcanic Exo-Ios that May Fuel Auroras on Super-Jupiters." 2026, *submitted to AAS journals*.
1. **Kotten, B.**, Soares-Furtado, M., Yarza, R.; et al. "Evaluating the Lithium Abundance of a Subgiant Star with a Brown Dwarf Companion: A Planetary Engulfment Candidate." 2026, *Accepted to The Astrophysical Journal*.

CO-AUTHOR PUBLICATIONS

6. Machol, J., Codrescu, S., Peck, C.; et al. **incl. Kotten, B.** "GOES-R Series X-Ray Sensor (XRS): 2. On-orbit measurements and calibrations." 2026, *Journal of Geophysical Research: Space Physics*, 131, e2026JA035181.
5. Limbach, M.A., Dacus, B., **Kotten, B.**; et al. "Exomoons and Exorings with the Habitable Worlds Observatory II: Finding 'Endor' with Lunar Eclipses." 2026, *The Astronomical Journal*, 171, 4.
4. Householder, A., Limbach, M. A., Biller, B., **Kotten, B.**; et al. "Sensitivity to Sub-Io-sized Exosatellite Transits in the MIRI LRS Lightcurve of the Nearest Substellar Worlds." 2025, *The Astrophysical Journal Letters*, 992, 1.

3. Wilson, M. J., Limbach, M. A., Skemer, A. J.; et al. **incl. Kotten, B.** "A Deep Search for Exomoons Around WISE 0855 With JWST." 2025, The Astronomical Journal, 170, 6.
2. Schulte, J., Rodriguez, J. E., Latham, D. W., Shields, J. V., Vowell, N., Soares-Furtado, M., **Kotten, B.**; et al. "Migration and Evolution of giant ExoPlanets (MEEP) II: Super-Jupiters and Lithium-rich Host Stars." 2025, Monthly Notices of the Royal Astronomical Society, 543, 1.
1. Vowell, N., Rodriguez, J. E., Latham, D. W., Quinn, S. N., Schulte, J.; et al. **incl. Kotten, B.** "Eleven New Transiting Brown Dwarfs and Very-low-mass Stars from TESS." 2025, The Astronomical Journal, 170, 2.

SCIENCE PRESENTATIONS

- Nov 2025 **Great Lakes Exoplanet Area Meeting**, Madison, WI [Talk]
 July 2025 **OWL Exoplanet Summer Program**, Santa Cruz, CA [Talk]
 June 2024 **244th American Astronomical Society Summer Meeting**, Madison, WI [Poster]
 Apr 2024 **UW-Madison Undergraduate Symposium**, Madison, WI [Poster]
 Jan 2024 **243rd American Astronomical Society Winter Meeting**, New Orleans, LA [Poster]
 Dec 2023 **UW-Madison Astronomy Club Symposium**, Madison, WI [Talk]
 Aug 2023 **Lamat REU Final Presentations 2023**, Santa Cruz, CA [Talk]
 Apr 2023 **UW-Madison Undergraduate Symposium**, Madison, WI [Talk]
 Apr 2023 **Space Weather Workshop**, Boulder, CO [Poster]
 Dec 2022 **UW-Madison Astronomy Club Symposium**, Madison, WI [Talk]
 July 2022 **Boulder Solar Alliance REU Final Presentations 2022**, Boulder, CO [Talk and Poster]
 Apr 2022 **UW-Madison Undergraduate Symposium**, Madison, WI [Talk]

AWARDS

- 2025 **Precandidate Rackham Graduate Student Research Grant**, University of Michigan
 2024 **Chambliss Undergraduate Poster Honorable Mention**, American Astronomical Society
 2024 **Graduate Research Fellowship Program**, National Science Foundation
 2024 **Lowell Doherty Award for Excellence in Astronomy**, UW-Madison Department of Astronomy
 2024 **Undergraduate Scholarship**, NASA Wisconsin Space Grant Consortium
 2023 **Bromley Conference Travel Award**, UW-Madison L&S Honors Program
 2023 **Dr. Maritza Irene Stapanian Crabtree Undergraduate Award**, UW-Madison
 2023 **Best General Radio News Story**, Wisconsin Broadcasters Assoc.
 2023 **Lamat Fellowship**, University of California Santa Cruz
 2021-2023 **Dean's List x3**, UW-Madison
 2022 **William F. Vilas Scholarship**, UW-Madison
 2021-2022 **Undergraduate Research Scholars**, UW-Madison
 2019-2021 **Dean's List x4**, Saint Cloud State University

COMPUTER SKILLS

LANGUAGES: Python, IDL, Matlab

PROJECTS:

GOES-R XRS (New) Flare Location Parameter Algorithm

May 2022-June 2023

Lead Developer and Maintainer

Improves the XRS flare location calculation algorithm with a Monte Carlo routine that optimizes the geometric factors in the algorithm via comparisons with locations determined using Solar Dynamics Observatory Atmospheric Imaging Assembly. Due to export control this project is not publicly available. Data and ReadMe available at <https://www.ngdc.noaa.gov/stp/satellite/goes-r.html>

Brown Dwarf Visualizer

August 2025

Co-Lead Developer

Created and released small package to plot brown dwarfs in an interactive plotting space during 2025 Code Astro program. See <https://github.com/astrojorge/BDViz>

TEACHING

UW-Madison Physics Learning Center, Peer Mentor Tutor

September 2023-May 2024

Group-based active learning for Physics 103, 104, 207, and 208

OBSERVATIONS

JWST Cycle 4 Program 8155, *Storms are Brewing in the ONC: Variability Monitoring of 50 Substellar Worlds* (43.3 hrs, PI Limbach; Co-I Kotten)

UW-Madison Pine Bluff 2.3 Meter Small Radio Telescope (ASTRON 465, 3 labs)

Peter Livingston Optical 16 inch Telescope (ASTRON 465, 2 labs)

OTHER PRESENTATIONS

- Mar 2026 **UMich Museum of Natural History, Ann Arbor, MI**
Women in Science & Engineering Residency Program invited speaker
- Dec 2025 **Astronomy on Tap, Ann Arbor, MI**
“Dear Pluto” (science play) performance
- Oct 2025 **University of Michigan, Ann Arbor, MI**
NSF Graduate Research Fellowships Program Q&A –Invited Panelist
- May 2024 **Astronomy on Tap, Madison, WI**
Dancing Lights: Exploring Auroras from Earth to Exoplanets
- Nov 2023 **Astronomy Club, Madison, WI**
Research Experience for Undergraduates and Internships Panel – Panelist
- July 2023 **Badger Talks Live, Madison, WI**
Devoured Worlds: Lessons From Planet-Ingesting Stars –Invited introduction
- Mar 2023 **WSUM News On Wisconsin Podcast, Madison, WI**
On Wisconsin podcast E20: Women in STEM interview –Invited speaker
- Dec 2022 **Astronomy Club, Madison, WI**
Research Experience for Undergraduates Panel – Panelist

ACTIVITIES, SERVICE, & OUTREACH

Rackham Student Government , Division II Representative	<i>May 2026-Present</i>
NSF GRFP Help Webpage (grfp-help.github.io), Creator and Maintainer	<i>February 2026-Present</i>
Astronomy on Tap Ann Arbor , Lead Organizer	<i>July 2025-Present</i>
University of Michigan Museum of Natural History , Planetarium Operator	<i>September 2024-Present</i>
UMich Astronomy Undergraduate Liason	<i>September 2025- September 2026</i>
ERES XI , LOC and Webmaster	<i>September 2025- July 2026</i>
UW-Madison Astronomy Department , Undergraduate Representative	<i>February 2023-May 2024</i>
UW-Madison Faculty Honors Committee , Student Representative	<i>September 2022-May 2024</i>
WSUM Student Radio News Team , Science Desk	<i>February 2022-February 2024</i>
UW-Madison Astronomy Club , Member	<i>September 2021-May 2024</i>